Shivam Balasaheb Bhagat

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PROFILE

MSc Advanced Computer Science student at the University of Manchester (graduating September 2026), with hands-on experience building backend systems, AI/ML applications, RAG pipelines, and full-stack web platforms. Strong in Python, JavaScript/TypeScript, REST APIs, PostgreSQL, Docker, and PyTorch. Seeking graduate software engineering, backend, AI/ML, or data engineering roles in the UK.

TECHNICAL SKILLS

Languages: Python, Java, JavaScript (ES6+), TypeScript **Front-end:** React, Next.js, HTML5, Tailwind CSS
Back-end / DB: Node.js, REST APIs, PostgreSQL, MySQL, MongoDB, Supabase
Data / ML: PyTorch, scikit-learn, Pandas, NumPy, OpenCV, Transformers, LangGraph, FAISS
Tools: Git/GitHub, Docker, VS Code, Linux, Vite, Vercel, Streamlit

EDUCATION

University of Manchester — MSc Advanced Computer Science *Sep 2025 – Sep 2026 (Expected)*

- Global Talent Scholarship recipient — awarded competitively to international students demonstrating exceptional academic potential.
- Focus areas: AI Applications, Fundamentals of Machine Learning, Secure Software Engineering, Secure System Design.
- Serving as Academic Representative, facilitating communication between students and faculty.

Savitribai Phule Pune University — B.E. Computer Engineering *Sep 2021 – Jun 2025*

- CGPA: 8.35/10 — First Class. Final-year project: Ask the Doc (Generative-AI document assistant).

PROJECTS

South Asian Culinary RAG Assistant (Group Project — 4 members)
LangGraph · FAISS · BGE-Large Embeddings · Qwen2.5 · Streamlit · Python

- Personal contributions: FAISS vector database construction, BGE-Large embedding pipeline, LangGraph intent-routing state machine, and 100-query evaluation benchmark.
- Designed a fully local RAG chatbot implementing a LangGraph state-machine with intent classification, semantic retrieval (1024-dim embeddings, L2 thresholding), and LLM-powered response generation.
- Achieved Recall@3 of 0.924, intent classification accuracy of 0.93, and generation faithfulness of 0.886 across seven intent categories on a structured evaluation set.
- Built a Streamlit dark-mode UI with session management, real-time streaming, and a pipeline-inspection panel exposing intent, extracted entities, and raw FAISS scores.

CI Pipeline Documentation Tool (MSc Dissertation — in progress)
Python · LLM APIs · YAML · Mermaid.js · Jinja2

- Building an automated tool that parses CI/CD pipeline configs (GitHub Actions, GitLab CI), generates structured natural-language summaries via LLM APIs, and renders pipeline-stage diagrams using Mermaid.js.
- Supervised by Dr. Suzanne Embury, University of Manchester; evaluating output quality using BLEU/ROUGE metrics against human-written documentation baselines.

Fashion-MNIST CNN Classifier — Constrained Deep Learning
PyTorch · Docker · Python

- Designed a compact 2-block CNN (50,378 parameters, sub-100K constraint) achieving 92.35% validation accuracy and 91.72% test accuracy on Fashion-MNIST.

- Conducted systematic ablation experiments across four configurations — baseline 2-CNN, CNN + Transformer hybrid (90.66%), data-augmented variant, and extended training schedule — selecting the optimal model based on generalisation.
- Trained and evaluated inside a Docker container without network access; applied ReduceLROnPlateau scheduling and BatchNorm for stable convergence within 15 epochs.

StoryWanderer — Web Novel Publishing Platform
React 18 · TypeScript · Vite · Tailwind CSS · shadcn/ui · Supabase
- Built a full-stack web application for publishing and reading serialised fiction; Supabase handles authentication, PostgreSQL storage, and real-time data sync.
- Implemented a component-driven frontend with React 18, TypeScript, and shadcn/ui; deployed on Vercel with chapter-level content management and a custom long-form reading UI.

Ask the Doc — Generative AI Document Assistant
Streamlit · HuggingFace Transformers · Pegasus · RoBERTa · gTTS · Python
- Built an end-to-end document intelligence app with abstractive summarisation (Pegasus) and extractive QA (RoBERTa-SQuAD2), integrated with text-to-speech output via gTTS.
- Final-year undergraduate project; deployed as a Streamlit application with a modular pipeline separating ingestion, inference, and rendering layers.

Real-Time Driver Drowsiness Detection
YOLOv5 · PyTorch · OpenCV · NumPy
- Developed a real-time fatigue detection system using a custom-trained YOLOv5 model classifying awake/drowsy eye states from live webcam feed at sustained frame rates.
- Implemented debounce logic to suppress false-positive alerts; trained on a custom labelled dataset optimised for variable lighting conditions.

LEADERSHIP & VOLUNTEERING

IEEE — Student Member & Event Organiser *Oct 2022 – Mar 2024*
- Co-organised technical workshops and AI/ML events; managed scheduling, venue coordination, and team logistics.

Volunteer — Saday Sadev NGO *Mar 2024 – Apr 2025*
- Produced social media content to increase programme awareness; drafted cross-organisation collaboration proposals.

National Service Scheme (NSS) — Member *Aug 2022 – Feb 2024*
- Organised a blood donation camp, supported community cleanliness drives, and volunteered at elderly care homes.

CERTIFICATIONS

ISC2 CC (Cybersecurity) 2024 · Python – NPTEL 2024 · Version Control – Coursera 2023 · Web Development – Udemy 2025